

CS 70 离散数学 和 概率论

笔记 14

春季 2026 Course Notes

条件的 概率, Independence, 和 Combinations of Events

One of the key properties of coin flips is independence:

if you flip a fair coin ten times and get ten T's, this

does not make it more likely that the next coin flip will be H's.

It still has exactly 50% chance of being H.

By contrast, 假设 当...时 dealing cards, the first ten cards are

all red (hearts 或 diamonds). What is the

chance that the next card is red?

We started with exactly 26 red cards and 26 black cards.

But after dealing

the first ten cards, we know that the deck has 16 red cards

和 26 black cards. 所以 the chance that the

next card is red is 16. 所以 unlike the case of coin flips,

now the chance of drawing a red card is no longer

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independent of the previous card that was dealt.

This is the phenomenon we will explore in this note.

1 条件的 概率

令's 考虑 an 例子 with a smaller 样本空间. 假设 we toss $m=4$ labeled

balls into $n=3$

labeled bins; this is a uniform sample space with $3^4=81$ sample points.

From the previous note, we already

know that the probability the first bin is empty is $(1-1/3)^4 = (2/3)^4 = 16/81$.

What is the probability of this event

$3^3 = 27$

given that the second bin is empty?

Let A denote the event that the first bin is empty, and B the event that the

second bin is empty.

In the language of conditional probability, we wish to compute the probability $P[A|B]$, which we read as "the conditional probability of A given B ."

How should we compute $P[A|B]$?

Since event B is guaranteed to happen, we need to look not at the whole

样本空间, 但是 at the smaller 样本空间 consisting only of the sample points in B . In terms of the

picture below, we are no longer looking at the large oval, but only the oval labeled B :

B

=

A

What should be the probability of each sample point ω

$\in B$ given that the event B occurs? If they all simply

inherited their probabilities from, then the sum of these probabilities would be $\sum$

$P[\omega] = P[B]$, 其中

$\omega \in B$

in general is less than 1.

所以, to get the correct normalization, we need to scale the probability of each sample

point by $1/P[B]$. That is, for each sample point ω

$\in B$, the new probability becomes

$P[\omega]$

$P[\omega]$

$P[\omega|B] = \frac{P[\omega]}{P[B]}$

$P[B]$

Now it is clear how to compute $P[A|B]$: 即,

we just sum up these scaled probabilities over all sample

points that lie in both A and B:
 $P[\omega] \quad P[A \cap B]$
 $P[A|B] = \frac{\sum P[\omega | B]}{\sum P[\omega]} = \frac{P[A \cap B]}{P[B]}$
 $\omega \in A \cap B \quad \omega \in A \cap B$
 CS70, Spring 2026, Note 14.1

定义 14.1 (条件的 概率). 对于 events A, B in the same 概率 space 满足 $P[B] > 0$, the conditional probability of A given B is

$P[A \cap B]$
 $P[A|B] = \frac{P[A \cap B]}{P[B]}$

Returning to our 例子 of balls and bins, to 计算 $P[A|B]$ we need to 图 out $P[A \cap B]$. 但是 $A \cap B$

is the 事件 that both the first two bins are empty, i.e., all four balls 秋季 in the third bin. 所以 $P[A \cap B] = \frac{1}{81}$

(why?). 因此,
 $P[A \cap B] = \frac{1}{81}$
 $P[A|B] = \frac{1/81}{16/81} = \frac{1}{16}$
 $P[B] = \frac{16}{81}$

非 surprisingly, $P[A|B] = \frac{1}{16} = 0.0625$ is quite a bit less than $P[A] = \frac{16}{81} \approx 0.1975$; knowing that bin 2 is

empty makes it significantly less likely that bin 1 will be empty.
 例子: Card Dealing

令' s apply the ideas discussed above to compute the probability that, when dealing 2 cards and the first card is known to be an ace, the second card is also an ace.
 Let B be the event that the first card is an ace, and let A be the event that the second card is an ace.
 To 计算 $P[A|B]$, we need to 图 out $P[A \cap B]$. This is the 概率 that both cards are aces. 笔记

that there are $52 \cdot 51$ sample points in the sample space, since each sample point is a sequence of two cards. A sample point is in $A \cap B$ if both cards are aces. This can happen in $4 \cdot 3 = 12$ ways. Since each sample point is equally likely, $P[A \cap B] = \frac{12}{52 \cdot 51}$, while $P[B]$, the probability of drawing an ace in

the first trial, is $\frac{4}{52}$. 因此,
 $P[A \cap B] = \frac{12}{52 \cdot 51}$
 $P[A|B] = \frac{12/52 \cdot 51}{4/52} = \frac{3}{51}$
 $P[B] = \frac{4}{52}$

其中 is less than $P[A] = \frac{4}{52} = \frac{1}{13}$ (see 练习 1). 因此, 如果 the first card is an ace, it makes it less likely

that the second card is also an ace.
 2 Bayesian Inference

Now that 我们有 introduced the notion of 条件的 概率, we can see how it is used in 实数 world settings. 条件的 概率 is at the heart of a subject called Bayesian inference, used extensively in fields such as machine learning, communications 和 signal processing. Bayesian inference is a way to update knowledge after making an observation. 对于 例子, we may have an estimate of the 概率 of a 给定 事件 A. After 事件 B occurs, we can update this estimate to $P[A|B]$. In this interpretation,

$P[A]$ can be thought of as a prior 概率: our assessment of the likelihood of an 事件 of interest, A , before making an observation. It reflects our prior knowledge. $P[A|B]$ can be interpreted as the posterior probability of A after the observation.

It reflects our updated knowledge.

Here is an example of where we can apply such a technique.

A pharmaceutical company is marketing a new test for a certain medical disorder.

According to clinical trials, the test has the following properties:

1.

When applied to an affected person, the test comes up positive in 90% of cases, and negative in 10% (these are called “false negatives”).

2. When applied to a healthy person, the test comes up 负的 in 80% of cases, and 正的 in 20% (these are called “false positives”).

CS70, Spring 2026, Note 14.2

Suppose that the incidence of the disorder in the US population is 5%; this is our prior knowledge.

When a

random person is tested and the test comes up 正的, how can we update the 概率 that the person

has the disorder? (Note that this is presumably not the same as the simple probability that a random person

in the population has the disorder, 其中 is just 1.) The implicit 概率 space here is the entire US

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population with the equal probabilities.

正的

$A \cap P$ $H \cap P$

Affected Healthy (P)

(A) (H)

负的 $A \cap N$ $H \cap N$

(N)

The 样本空间 here consists of all people in the US—表示 their number by N (所以 $N \approx 350$ million).

令 A be the 事件 that a person chosen at random is affected, and

B be the 事件 that a person chosen at

random tests positive. Now we can rewrite the information above as:

- $P[A] = 0.05$, (5% of the U.S. population is affected)
- $P[B|A] = 0.9$, (90% of the affected people test positive)
- $P[B|A] = 0.2$, (20% of healthy people test positive)

We want to calculate $P[A|B]$. We can proceed as follows:

$P[A \cap B]$ $P[B|A]P[A]$

$P[A|B] = \frac{P[A \cap B]}{P[B]}$ (1)

$P[B]$ $P[B]$

We obtained these two equalities above by applying the definition of conditional probability:

$P[A \cap B]$

$P[B|A] = \frac{P[A \cap B]}{P[A]}$

$P[A]$

To evaluate (1), we need to 计算 $P[B]$. This is the 概率 that a random person tests 正的. To

计算 this, we can sum two values: the 概率 $P[A \cap B]$ that a healthy person tests 正的 and the

概率 $P[A \cap B]$ that an affected person tests 正的. We can sum because the events $A \cap B$ and $A \cap B$

do not intersect:

B

A

$A \cap B$ $A \cap B$

By again applying the definition of conditional probability, we have:

$P[B] = P[A \cap B] + P[A \cap B] = P[B|A]P[A] + P[B|A]P[A]$

(2)

$= P[B|A]P[A] + P[B|A](1 - P[A])$.

CS70, Spring 2026, Note 14.3

Combining (1) and (2), we have expressed $P[A|B]$ in terms of $P[A]$, $P[B|A]$ and $P[B|A]$:

$P[B|A]P[A]$

$P[A|B] = \dots$ (3)

$P[B|A]P[A] + P[B|A](1 - P[A])$

By plugging in the values written above, we obtain $P[A|B] = \frac{9}{10} \approx 0.19$.

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方程 (3) is useful 对于 many inference problems. We are 给定 $P[A]$,

其中 is the (unconditional) prob-

ability that the event of interest, A, happens.

We are given $P[B|A]$ and $P[B|A]$, which quantify how noisy

the observation is. (如果 $P[B|A]=1$ 和 $P[B|A]=0$, 对于 例子, the observation is completely noiseless.)

Now we want to calculate $P[A|B]$, the 概率 that the 事件 of interest happens 给定 we made the observation. 方程 (3) allows us to do just that.

Of course, (1) to (3) are derived from the basic axioms of

概率 and the 定义 of 条件的

概率, and are 因此 真 with 或 without the above Bayesian inference interpretation. 然而,

this interpretation is very useful when we apply probability theory to study inference problems.

3 贝叶斯' Rule 和 Total 概率 Rule

Equations (1) and (2) are very useful in their own right. The

former is called 贝叶斯' Rule and the latter is

called the Total 概率 Rule. 贝叶斯' Rule is useful when one wants to calculate $P[A|B]$ 但是 one is

given $P[B|A]$ instead, i.e., it allows us to "flip things around."

The Total 概率 Rule is an application of the strategy of

"dividing into cases." 存在 two possi-

bilities: either an 事件 A happens 或 A does 非 happen. 如果 A happens, the 概率 that B happens is

$P[B|A]$. If A does not happen, the probability that B happens is $P[B|A]$.

If we know or can easily calculate

these two probabilities and also $P[A]$, then the Total Probability Rule yields the probability of event B.

3.1 Examples

Tennis Match

You are about to play a tennis match against a randomly chosen opponent 和 you wish to calculate your probability of winning.

You know your opponent will be one of two people, X or Y. If person X is chosen,

you will win with probability 0.7. If person Y is chosen,

you will win with probability 0.3. Your opponent

is chosen by flipping a biased coin such that the probability of choosing X is 0.6.

令' s first determine which events we are interested in.

Let A be the event that you win. Let B be the event

X

that person X is chosen, and let B be the event that person Y is chosen.

We wish to calculate $P[A]$. Here is

Y
 what we know so far:
 • $P[A|B] = 0.7$, (if person X is chosen, you win with probability 0.7)
 X
 • $P[A|B] = 0.3$, (if person Y is chosen, you win with probability 0.3)
 Y
 • $P[B] = 0.6$, (person X is chosen with probability 0.6)
 X
 • $P[B] = 0.4$, (person Y is chosen with probability 0.4)
 Y
 By using the Total Probability Rule, we have $P[A] = P[A|B]P[B] + P[A|B]P[B]$, and plugging in the
 X X Y Y
 known values gives $P[A] = (0.7 \times 0.6) + (0.3 \times 0.4) = 0.54$.
 CS70, Spring 2026, Note 14.4

Balls and Bins

Imagine we have the following two bins each containing some number of black and white balls:

Bin1 Bin2

假设 one of the two bins is chosen with equal 概率 and a ball is drawn from the chosen bin uniformly at random. What is the 概率 that we picked Bin 1 给定 that a white ball was drawn, i.e.,

$P[\text{Bin1} |]$?

Is the 答案 2, 因为 we know that 存在 a total of three white balls, two of 其中 are in Bin 1? This

3
 reasoning is incorrect.

Instead, what we should do is appropriately scale each sample point as the following picture shows:

B B

1 2

1 1 1 1

10 10 10 4

1 1 1

10 10 4

This diagram shows that the sample space consists of the outcomes in event B (corresponding to Bin1)

1
 and event B (corresponding to Bin2), i.e., $\Omega = B \cup B$.

We can use the definition of conditional probability

$2 \mid 2$

to see that

$1 + 1 \mid 2 \mid 4$

$P[\text{Bin1} |] = \frac{10}{10+10} = \frac{1}{2}$.

$1 + 1 + 1 \mid 9 \mid 9$

$10 \mid 10 \mid 4 \mid 20$

令 us try to achieve this 概率 using 贝叶斯' Rule. To apply 贝叶斯' Rule, we need to 计算

$P[\mid \text{Bin1}]$, $P[\text{Bin1}]$, and $P[]$. Here, $P[\mid \text{Bin1}]$ is the chance that we pick a white ball 给定 that we picked Bin1, which is $\frac{2}{10}$. $P[\text{Bin1}] =$

$\frac{1}{2}$, as given in the description of the problem. Finally, $P[]$ can be

$5 \mid 2$

computed using the Total Probability Rule:

$2 \mid 1 \mid 1 \mid 1 \mid 9$

$P[] = P[\mid \text{Bin1}] \times P[\text{Bin1}] + P[\mid \text{Bin2}] \times P[\text{Bin2}] = \frac{1}{2} \times \frac{1}{2} + \frac{1}{2} \times \frac{1}{2} = \frac{1}{2}$.

$5 \mid 2 \mid 2 \mid 2 \mid 20$

Observe that we can apply the Total Probability Rule here because $P[\text{Bin1}]$ is the complement of $P[\text{Bin2}]$. Finally, upon plugging the above values into 贝叶斯' Rule, 我们得到 the 概率 that we picked Bin 1 given that we picked a white ball:

$$P[\text{Bin1} | \text{white}] = \frac{2 \times \frac{1}{20}}{\frac{2}{20} + \frac{1}{20}} = \frac{2}{3} = \frac{2}{3}.$$

All 我们有 done above is to combine 贝叶斯' Rule 和 the Total 概率 Rule; this is also how we obtained (3). Equivalently, we could have plugged in the appropriate values to (3). CS70, Spring 2026, Note 14.5

3.2 Generalization

We now 考虑 贝叶斯' Rule 和 the Total 概率 Rule in a more general context. First, we 定义 a partition of an event as follows.

Definition 14.2 (Partition of an event).

We say that an event A is partitioned into events $A_1, \dots, A_n$ 如果

1. $A = A_1 \cup A_2 \cup \dots \cup A_n$, 和
2. $A_i \cap A_j = \emptyset$ for all $i \neq j$ (i.e., $A_1, \dots, A_n$ are mutually exclusive).

In other words, each outcome in A belongs to exactly one of the events $A_1, \dots, A_n$.

Now, let $A_1, \dots, A_n$ be a partition of the sample space Ω . 那么, the Total Probability Rule for any event B

is

$$P[B] = \sum_{i=1}^n P[B \cap A_i] = \sum_{i=1}^n P[B|A_i]P[A_i], \quad (4)$$

while Bayes' Rule, assuming $P[A_i] > 0$, is given by

$$P[A_i | B] = \frac{P[B|A_i]P[A_i]}{\sum_{j=1}^n P[B|A_j]P[A_j]}, \quad (5)$$

where these two equalities follow from the Total Probability Rule.

4 Combinations of Events

In most applications of 概率 in Computer Science, we are interested in things like $P[(A_1 \cap \dots \cap A_n) | B]$ 和

$P[(A_1 \cup \dots \cup A_n) | B]$, 其中 the A_i are simple events (i.e., we know or can easily 计算 $P[A_i]$). The 交集

$A_1 \cap \dots \cap A_n$

corresponds to the logical AND of the events $A_1, \dots, A_n$, while the union $A_1 \cup \dots \cup A_n$

corresponds to their logical OR.

As an 例子, 如果 A denotes the 事件 that a failure of type i happens in a certain system, 那么 (cid:83) A is the i i i event that the system fails.

In general, computing the probabilities of such combinations can be very difficult. In this section, we discuss some situations where it can be done.

令' s start with independent events, for which intersections are quite simple to compute.

4.1 独立的 Events

定义 14.3 (Independence). Two events A, B in the same 概率 space are said to be 独立的

如果 $P[A \cap B] = P[A] \times P[B]$.

The intuition behind this definition is the following.

Suppose that $P[B] > 0$. Then we have

$$P[A \cap B] = P[A] \times P[B]$$

$$P[A|B] = P[A].$$

$$P[B|A] = P[B]$$

因此 independence has the natural meaning that “the 概率 of A is not affected by whether or not B occurs.”

(By a symmetrical argument, we also have $P[B|A] = P[B]$ provided $P[A] > 0$.)

For events A, B such

that $P[B] > 0$, the condition $P[A|B] = P[A]$ is actually equivalent to the definition of independence.

Several of our previously mentioned random experiments consist of independent events.

For example, if we

flip a coin twice, the event of obtaining heads in the first trial is independent of the event of obtaining heads in the second trial.

The same applies for two rolls of a die; the outcomes of each trial are independent.

CS70, Spring 2026, Note 14.6

The above definition generalizes to any finite set of events:

定义 14.4 (Mutual independence). Events $A_1, \dots, A_n$ are said to be mutually independent 如果 对于 every

$I \subseteq \{1, \dots, n\}$

subset $I \subseteq \{1, \dots, n\}$ with size $|I| \geq 2$,

$$P[(cid:84) A_I] = \prod_{i \in I} P[A_i]. \quad (6)$$

$i \in I$ $i \in I$

$i \in I$

Anequivalent definition of mutual independence is as follows.

定义 14.5 (Mutual independence). Events $A_1, \dots, A_n$ are said to be mutually independent 如果 对所有

$1 \leq i \leq n$

$B_i \in \{A_i, \Omega\}, i = 1, \dots, n,$

$i \in I$ $i \in I$

n

$$P[B_1 \cap \dots \cap B_n] = \prod_{i=1}^n P[B_i]. \quad (7)$$

$1 \leq i \leq n$

$i=1$

Remarks.

1. In 定义 14.4, (6) needs to hold 对于 every 子集 I of $\{1, \dots, n\}$ with size $|I| \geq 2$. The cases of

$|I|=0$ and $|I|=1$ are omitted, as they impose no constraints.

2.

Note that (6) imposes $2^n - n - 1$ constraints on the probability distribution, 当...时 (7) defines $2^n - n$ constraints.

straints.

It turns out that exactly $n+1$ constraints implied by (7) are actually redundant.

For mutually independent events $A_1, \dots, A_n$,

it is not hard to check from the definition of conditional probability that,

for any

$1 \leq i \leq n$ and any subset $I \subseteq \{1, \dots, n\} \setminus \{i\}$, we have

$P[A_i | (A_j)_{j \in I}] = P[A_i]$.

$i, j \in I$

笔记: that the independence of every pair of events (所以-called pairwise independence) does not necessarily

imply mutual independence.

As illustrated in the following example, it is possible to construct three events

A, B, C such that each pair is independent but the triple A, B, C is not mutually independent.

例子: Pairwise Independent but Not Mutually Independent Events

假设 you toss a fair coin twice and let A be the event that the first flip is H and B be the event that the

second flip is H. Now let C

be the event that both flips are the same (i.e., both H's or both T's).

Of course

A and B are independent. What is more interesting is that A and

C : 给定 that the first toss came

up H, the chance of the second flip being the same as the first is still 1. Another way of saying this is that

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$P[A \cap C] = P[A]P[C]$ 因为 $A \cap C$ is the event that the first flip is H and the second is also H. By the

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same reasoning B and C are also independent.

On the other hand, A, B and C are not mutually independent.

For example, if we are given that A and B occurred, then the probability that C occurs is 1.

所以, even though

A, B and C are not mutually independent, every pair of them are independent.

换句话说, A, B and C

are pairwise independent but not mutually independent.

4.2 Intersections of Events

Computing the probability of an intersection of mutually independent events is easy; it follows from the

definition. We simply multiply the probabilities of each event. How do

we compute the probability of an

intersection for events that are not mutually independent? From the definition of conditional probability,

we immediately have the following Product Rule (sometimes

also called the chain rule) for computing the

probability of an intersection of events.

CS70, Spring 2026, Note 14.7

Theorem 14.1 (Product Rule). For any events A, B , we have

$P[A \cap B] = P[A]P[B|A]$.

More generally, for any events $A_1, \dots, A_n$,

for any

$P[(A_1 \cap \dots \cap A_n) | (A_1 \cap \dots \cap A_{i-1})] = P[A_i | A_1 \cap \dots \cap A_{i-1}]$.

$i=1, \dots, n$

证明.

The first assertion follows directly from the definition of $P[B|A]$ (and is in fact a special case of the second assertion with $n=2$).

To prove the second assertion, we will use induction on n (the number of events). The base case is $n=1$,

and corresponds to the statement that $P[A] = P[A]$, which is

trivially true. For an arbitrary $n > 1$, assume

(the inductive hypothesis) that

$$P[(\text{cid:84})n-1A] = P[A] \times P[A|A] \times \cdots \times P[A|(\text{cid:84})n-2A].$$

Now we can apply the definition of conditional probability to the two events A and $(\text{cid:84})n-1A$ to deduce that

$$\begin{aligned} P[(\text{cid:84})nA] &= P[A \cap ((\text{cid:84})n-1A)] \\ &= P[A | (\text{cid:84})n-1A] \times P[(\text{cid:84})n-1A] \\ &= P[A | (\text{cid:84})n-1A] \times P[A] \times P[A|A] \times \cdots \times P[A | (\text{cid:84})n-2A] \end{aligned}$$

where in the last line we have used the inductive hypothesis.
 This completes the proof by induction.

The Product Rule is particularly useful when we can view our sample space as a sequence of choices.

The next few examples illustrate this point.

例子: Coin Tosses

Toss a fair coin three times.

Let A be the event that all three tosses are heads. Then $A = A_1 \cap A_2 \cap A_3$, 其中

A_i

is the event that the i th toss comes up heads. We have

$$P[A] = P[A_1] \times P[A_2 | A_1] \times P[A_3 | A_1 \cap A_2]$$

$= \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}$

$$= \frac{1}{2^3} = \frac{1}{8}.$$

The second line here follows from the fact that the tosses are mutually independent.

Of course, we already

know that $P[A_i] = \frac{1}{2}$

from our definition of the probability space in the previous note.

Another way of looking

at this calculation is that it justifies our definition of the probability space,

and shows that it was consistent

with assuming that the coin flips are mutually independent.

If the coin is biased with heads probability

p , we get, again using independence,

$$P[A] = P[A_1] \times P[A_2] \times P[A_3] = p^3.$$

More generally, the probability of any sequence of n tosses containing k heads and $n-k$ tails is $p^k(1-p)^{n-k}$.

This is in fact the reason we defined the probability space this way:

we defined the sample point probabilities

so that the coin tosses would behave independently.

CS70, Spring 2026, Note 14.8

例子: Monty Hall Revisited

Recall the Monty Hall 题目 from the previous 笔记: 存在 three doors 和 the 概率 that the

prize is behind any given door is $\frac{1}{3}$.

There are goats behind the other two doors. The contestant picks a door

randomly, and the host opens one of the other two doors, revealing a goat.

How do we calculate intersections

in this setting? 对于例子, what is the 概率 that the contestant chooses door 1, the prize is behind

door 2, and the host chooses door 3?

Let C be the event that the contestant chooses door i , let P

be the event that the prize is behind door i , 和

i i

let H be the event that the host chooses door i . 那么, by the Product Rule,

i

$P[C \cap P \cap H] = P[C] \times P[P | C] \times P[H | C \cap P]$.

1 2 3 1 2 1 3 1 2

The probability of C is

1, since the contestant is choosing the door at random.

The probability of P 给定

1 3 2

C is still 1 since they are independent.

The probability of the host choosing door 3 given events C and P

1 3 1 2

is 1; the host cannot choose door 1 因为 the contestant has already chosen it, 和 the host cannot choose

door 2 since the host must reveal a goat (and not the prize). 因此,

1 1 1

$P[C \cap P \cap H] = \frac{1}{3} \times 1 \times 1 = \frac{1}{3}$.

1 2 3

3 3 9

观察到 we needed 条件的 概率 in this setting; had we simply multiplied the probabilities

of each 事件, we would have obtained 1 因为 the 概率 of H is also

1 (can you 图 out why?).

27 3 3

What 如果 we changed the situation, 和 instead asked 对于 the 概率 that the contestant chooses door 1,

the prize is behind door 1, 和 the host chooses door 3? We

can use the same technique as above, 但是 our

final answer will be different.

This is left as an exercise (see Exercise 4).

Now, noting that $P[H | C] = 1$ (see Exercise 5) and $P[C \cap H] = P[C] P[H$

$| C] = 1 \times 1 = 1$, we obtain

3 1 2 1 3 1 3 1 3 2 6

$P[C \cap P \cap H] = \frac{1}{2}$

$P[P | C \cap H] = \frac{1}{2} \times \frac{2}{3} = \frac{1}{3}$,

2 1 3 $P[C \cap H] = \frac{1}{3}$

1 3

6

which is a more formal derivation of the same conclusion described in the previous note.

例子: Poker Hands

令' us use the Product Rule to compute the probability of a flush in a different way.

This is equal to $4 \times P[A]$,

where A is the probability of a Hearts flush.

Intuitively, this should be clear since there are 4 suits; we' ll see

why this is formally 真 in the next 节. We can write

$A = \bigcup_{i=1}^4 A_i$, 其中 A_i is the 事件 that the i th

$i=1$ i i

card we pick is a Heart. So we have
 $P[A] = P[A_1] \times P[A_2 | A_1] \times \cdots \times P[A_5 | (A_1, \dots, A_4)]$.
 $1/2 \times 1/5 = 1/10$
Clearly $P[A_1] = 1/2$. What about $P[A_2 | A_1]$? Well, 因为 we
are conditioning on A_1 (the first card
 $1/5 \times 4/4 = 1/5$
is a Heart), 存在 only 4 remaining possibilities 对于 the
second card, 4 of 其中 are Hearts. 所以
 $P[A_2 | A_1] = 4/5$. Similarly, $P[A_3 | A_1 \cap A_2] = 3/4$, and soon. So we get
 $1/2 \times 4/5 \times 3/4 \times 2/3 \times 1/2 = 1/10$
 $4 \times P[A] = 4 \times 1/10 = 2/5$,
 $5/2 \times 1/5 = 1/2$
which is exactly the same fraction we computed in the previous note.

So now we have two methods of computing probabilities in many of our sample spaces.
It is useful to keep
these different methods around, both as a check on your
answers 和 because in some cases one of the
methods is easier to use than the other.
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4.3 Unions of Events

You are in Las Vegas, 和 you spy a new game with the
following rules. You pick a number between 1
and 6. Then three dice are thrown.
You win if and only if your number comes up on at least one of the dice.

The casino claims that your odds of winning are 50%, using the following argument.
Let A_i be the event that
you win. We can write $A = A_1 \cup A_2 \cup A_3$, where A_i
is the event that your number comes up on die i . Clearly

$1/6$
 $P[A_i] = 1/6$ for each i . 因此, they claim
 $1/6$
 $1/6$
 $P[A] = P[A_1 \cup A_2 \cup A_3] = P[A_1] + P[A_2] + P[A_3] = 3 \times 1/6 = 1/2$.
 $1/2 \times 3 = 1/2$
 $6/2 = 3$

Is this calculation correct? Well, 假设 instead that the
casino rolled six dice, 和 again you win 如果 和
only 如果 your number comes up at least once. 那么 the analogous
calculation would say that you win with
probability $6 \times 1/6 = 1$, i.e., certainly!

The situation becomes even more ridiculous when the number of dice
6
gets bigger than 6.

The problem is that the events A_i are not disjoint:

i.e., there are some sample points that lie in more than one
 A_i
of the A_i .

(We could get really lucky and our number could come up on two of the dice, or all three.)

所以, 如果

$1/6$
we add up the $P[A_i]$, we are counting some sample points more than once.
 $1/6$

Fortunately, in Note 12 we learned about the Principle of Inclusion-Exclusion which allows us to deal with
this kind of situation. In the 证明 of 定理 12.3, it was shown
that every $\omega \in A_1 \cup \cdots \cup A_n$ is 非

$1 \leq n$
 counted by the Inclusion-Exclusion formula, 当...时 every $\omega \in A$
 $\cup \dots \cup A$ is counted exactly once by the
 $1 \leq n$
 formula. 因此, rather than summing (with appropriate signs)
 the cardinalities of (cid:84) A in the Inclusion-
 $i \in S$ i
 Exclusion formula, if we instead sum their probabilities $P[(cid:84)$
 $A]$, we obtain the following useful formula
 $i \in S$ i
 for computing $P[A \cup \dots \cup A]$:
 $1 \leq n$
 Theorem 14.2 (Inclusion-Exclusion). Let $A, \dots, A$
 be events in some probability space, where $n \geq 2$. 那么,
 $1 \leq n$
 we have
 n
 $P[A \cup \dots \cup A] = \sum_{k=1}^n \binom{n}{k} (-1)^{k+1} \sum_{i \in S, |S|=k} P[(cid:84) A]. \quad (8)$
 $1 \leq n$ $i \in S$ i
 $k=1$ $S \subseteq \{1, \dots, n\}: |S|=k$
 The right hand side of (8) can be written as
 n
 $P[(cid:83) \sum_{i=1}^n P[A] - \sum_{i=1}^n \sum_{j=1}^n P[A \cap A] + \sum_{i=1}^n \sum_{j=1}^n \sum_{k=1}^n P[A \cap A \cap A]$
 $] \dots + (-1)^{n+1} P[A \cap \dots \cap A],$
 $i=1$ $i < j$ $i < j < k$
 其中 $\sum$ denotes summing over all $i, j \in \{1, \dots, n\}$ 满足 $i < j$, 和 所以
 on. 即, to 计算
 $i < j$
 $P[(cid:83)$
 $A]$, we start by summing the event probabilities $P[A]$, then we subtract the probabilities of all pairwise
 $i < j$ i

intersections, then we add back in the probabilities of all three-way intersections, and soon.
 You might like
 to verify it for the special case $n=3$ by drawing a Venn diagram.
 You might also like to prove the formula

for general n by induction (in similar fashion to the proof of the Product Rule above).

Using (8), the probability we get lucky in the new game in Las Vegas is given by
 $P[A \cup A \cup A] = P[A] + P[A] + P[A] - P[A \cap A] - P[A \cap A] - P[A \cap A] + P[A$
 $\cap A \cap A]$.
 $1 \leq 2 \leq 3$ $1 \leq 2 \leq 3$ $1 \leq 2 \leq 3$ $1 \leq 2 \leq 3$ $1 \leq 2 \leq 3$
 Now the nice thing here is that the events A are mutually
 独立的 (the 结果 of any die does 非
 i
 depend on that of the others), 所以 $P[A \cap A] = P[A]P[A] =$
 $(1/2)^2 = 1/4$, 和 similarly $P[A \cap A \cap A] =$
 $(1/2)^3 = 1/8$. 所以, we get
 $6 \times 1/4$
 $(cid:18) (cid:19) (cid:18) (cid:19)$
 $1 \leq 1 \leq 1$ 91
 $P[A \cup A \cup A] = 3 \times \frac{1}{2} - 3 \times \frac{1}{4} + \frac{1}{8} = \approx 0.42.$
 $1 \leq 2 \leq 3$
 $6 \times 36 \times 216 \times 216$
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Soyouroddsarequiteabitworse than what the casino is claiming!
 When n is large (i.e., we are interested in the 并集 of many events), the Inclusion-Exclusion formula is essentially useless because it involves computing the 概率 of the 交集 of every non-empty 子集 of the events: 和 存在 $2^n - 1$ of these! Sometimes we can just look at the first few terms of it 和 forget the rest: 笔记 that summing the first k terms gives us an overestimate when k is odd 和 an underestimate when k is even, and both these estimates get better as k increases.

然而, in many situations we can get along way by just looking at the first term:

1. (Mutually exclusive events) 如果 the events $A_1, \dots, A_n$ are mutually exclusive (i.e., $A_i \cap A_j = \emptyset$ 对所有 $i \neq j$), 那么

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n P[A_i].$$

[笔记 that 我们有 already used this fact several times in our examples, e.g., in claiming that the

probability of a flush is four times the probability of a hearts flush—clearly flushes in different suits are disjoint events.]

2. (Union bound) Let $A_1, \dots, A_n$ be events in some probability space.

那么, for all $n \in \mathbb{Z}^+$,

$$P\left(\bigcup_{i=1}^n A_i\right) \leq \sum_{i=1}^n P[A_i]. \quad (9)$$

This merely says that adding up the $P[A_i]$ can only overestimate the probability of the union. Crude as it

may seem, we will later see how to use the union bound effectively in Computer Science examples.

5 Exercises

1. 假设 five cards are dealt from a standard deck of playing cards. 令 A_i be the 事件 that the i th

card is an ace. Show that $P[A_i] = 1/5$ for all $i = 1, \dots, 5$.

2. Prove Equations (4) and (5).

3. Show that Definitions 14.4 and 14.5 of mutual independence are equivalent.

4. In the Monty Hall problem, find $P[C \cap H]$.

5. In the Monty Hall problem, show that $P[H|C] = 1/3$.

6. Prove the union bound from (9).

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